Angle Pairs Formed by a Transversal
Answer Key

High School Geometry — Lines and Angles

Quick Answers

1. —	3. 65, 115	5. 74, 106	7. 30, 111
2. 112, 68	4. 17, 61	6. 25, 65	8. —

Curriculum

Level: High School Geometry — Lines and Angles
HSG-CO.A–HSG-CO.D – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 14 tagged checks.

How to grade this. The pair *name* is half of each answer. A correct measure with the wrong name ("vertical" where the student used corresponding angles) shows the arithmetic worked and the reasoning did not; mark the name. Angle numbering follows the sheet's diagram: angles 1–4 around the upper intersection (1 upper right, 2 upper left, 3 lower left, 4 lower right) and 5–8 around the lower one, same order.

1. Name each pair: $\angle 1$ and $\angle 5$; $\angle 4$ and $\angle 6$; $\angle 2$ and $\angle 8$.
Solution: Read position, not size.

- $\angle 1$ and $\angle 5$ sit in the *same* corner at their own intersections (upper right at each): **corresponding angles**.
- $\angle 4$ and $\angle 6$ are both between the parallel lines and on *opposite* sides of t: **alternate interior angles**.
- $\angle 2$ and $\angle 8$ are both outside the parallel lines and on opposite sides of t: **alternate exterior angles**.

All three pairs are congruent pairs, because $\ell_1 \parallel \ell_2$.

2. $m\angle 1 = 112°$. Find $m\angle 5$, then $m\angle 6$.
Solution: $\angle 1$ and $\angle 5$ are corresponding angles and the lines are parallel, so they are congruent: $\boxed{m\angle 5 = 112°}$.
Next, $\angle 5$ and $\angle 6$ are adjacent along ℓ_2, a linear pair, so they are supplementary: $180° - 112° = \boxed{m\angle 6 = 68°}$.

3. $m\angle 4 = 65°$. Find $m\angle 6$, then $m\angle 5$.
Solution: $\angle 4$ and $\angle 6$ are alternate interior angles (both between the lines, opposite sides of t), so they are congruent: $\boxed{m\angle 6 = 65°}$.
$\angle 4$ and $\angle 5$ are same-side interior angles (both between the lines, same side of t), so they are supplementary rather than congruent: $180° - 65° = \boxed{m\angle 5 = 115°}$. The two names are what decide congruent versus supplementary.

4. Corresponding angles: $m\angle 2 = (3x + 10)^\circ$, $m\angle 6 = (5x - 24)^\circ$.
Solution: Corresponding angles on parallel lines are congruent, so set the expressions equal — that is the equation the pair name justifies.

$$\begin{aligned} 3x + 10 &= 5x - 24 \\ 34 &= 2x \\ x &= 17 \end{aligned}$$

So $\boxed{x = 17}$, and substituting back, $m\angle 2 = 3(17) + 10 = \boxed{61^\circ}$. Check the partner: $5(17) - 24 = 61$ as well.

5. $m\angle 7 = 74^\circ$. Find $m\angle 1$, then $m\angle 4$.
Solution: $\angle 7$ and $\angle 1$ are alternate exterior angles (both outside the parallel lines, opposite sides of t), so they are congruent: $\boxed{m\angle 1 = 74^\circ}$.
$\angle 1$ and $\angle 4$ are adjacent at the upper intersection — a linear pair along t — so they are supplementary: $180^\circ - 74^\circ = \boxed{m\angle 4 = 106^\circ}$.

6. Same-side interior: $m\angle 3 = (2x + 15)^\circ$, $m\angle 6 = (3x + 40)^\circ$.
Solution: Same-side interior angles on parallel lines are supplementary, so the expressions *sum* to 180; setting them equal is the standard error here.

$$\begin{aligned} (2x + 15) + (3x + 40) &= 180 \\ 5x + 55 &= 180 \\ x &= 25 \end{aligned}$$

So $\boxed{x = 25}$ and $m\angle 3 = 2(25) + 15 = \boxed{65^\circ}$. Check: $m\angle 6 = 3(25) + 40 = 115$, and $65 + 115 = 180$.

7. Alternate exterior: $m\angle 2 = (4x - 9)^\circ$, $m\angle 8 = (3x + 21)^\circ$.
Solution: Alternate exterior angles are congruent, so

$$\begin{aligned} 4x - 9 &= 3x + 21 \\ x &= 30 \end{aligned}$$

giving $\boxed{x = 30}$ and $m\angle 2 = 4(30) - 9 = \boxed{111^\circ}$. Second check: $m\angle 8 = 3(30) + 21 = 111$ too, and $\angle 7$ is the linear pair partner of $\angle 8$ along ℓ_2, so $m\angle 7 = 180 - 111 = 69$.

8. Explain why $\angle 3$ and $\angle 6$ are supplementary when $\ell_1 \parallel \ell_2$. *(Manual review — accept any correct chain.)*
Model answer: $\angle 3$ and $\angle 5$ are alternate interior angles, so with $\ell_1 \parallel \ell_2$ they are congruent: $m\angle 3 = m\angle 5$. Angles 5 and 6 are adjacent along ℓ_2, a linear pair, so $m\angle 5 + m\angle 6 = 180$. Substituting the first statement into the second gives $m\angle 3 + m\angle 6 = 180$, which is what supplementary means.
An equally good chain runs through corresponding angles: $\angle 3$ and $\angle 7$ are corresponding, hence congruent, and $\angle 7$ and $\angle 6$ form a linear pair. What must be present for full credit is a named congruent pair, a named supplementary pair, and the substitution that joins them — "they look supplementary" is not a reason.